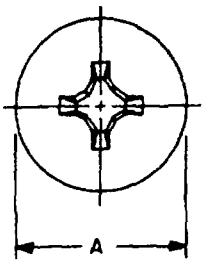
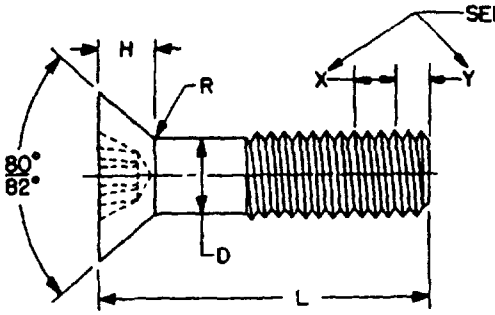
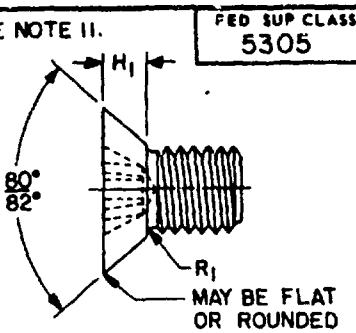
FED SUP CLASS 5305





FULL HEAD



UNDERCUT HEAD

TABLE 1

NOMINAL SIZE			(#0) .060	(#1) 073	(#2) .086	(#4) .112	(#6) .138	(#8) .164	(#10) .190	(1/4) .250	(5/16) .3125	(3/8) .375	(7/16) .4375	(1/2) .500
THREADS PER INCH UNF-2A			80	72	64	48	40	36	32	28	24	24	20	20
D BODY DIAMETER		MAX	0600	0730	0860	.1120	.1380	.1640	1900	2500	.3125	.3750	.4375	.5000
		MIN	0496	0615	.0733	.0954	.1184	.1424	.1658	.2225	.2806	.3430	.3995	.4619
A HEAD DIA		MAX SHARP	.119	146	.172	225	279	.332	.385	.507	.635	.762	.812	.875
		MIN SHARP	.105	130	.156	.207	.257	.308	359	.477	.600	.722	.767	.831
		ABS MIN	099	123	.147	.195	.244	.292	.340	.452	.568	.685	.723	.775
H HEAD HEIGHT		REF	.035	.043	051	.067	.083	.100	.116	.153	.191	.230	.223	.223
H_1 HEAD HEIGHT-UNDERCUT		MAX	.025	.031	036	.047	059	.070	.081	.107	.134	.161	.156	.156
		MIN	.018	.023	.028	.038	.048	.058	.068	.092	.116	.140	.133	.130
R RADIUS		MAX	.024	.029	.034	.045	.055	.066	.076	.100	.125	.150	.175	.200
R_1 RADIUS-UNDERCUT		MAX	.009	.011	.013	.017	.021	025	.029	.038	.047	.056	.066	.075
TENSILE STRENGTH LBS-MIN			105	165	235	395	605	880	1,200	2,180	3,480	5,260	7,120	9,590
L LENGTH	TOL		DASH NO.	DASH NO.	DASH NO	DASH NO.	DASH NO.	DASH NO	DASH NO.	DASH NO.	DASH NO.	DASH NO	DASH NO.	DASH NO.
.125			201	206*	213	222	233	248						
188			202	207*	214	223	234	249	267					
.250			203	208*	215	224	235	250	268	285				
.312			204	209*	216	225	236	251	269	286				
.375	+0		205	210*	217	226	237	252	270	287	302			
.438	-.031			211*	218	227	238	253	271	288	303			
.500				212*	219	228	239	254	272	289	304	317		
.625					220	229	240	255	273	290	305	318	330*	
750					221	230	241	256	274	291	306	319	331*	342
875						231	242	257	275	292	307	320	332*	343
1 000						232	243	258	276	293	308	321	333*	344
1.250							244	259	277	294	309	322	334*	345
1.500	+0						245	260	278	295	310	323	335*	346
1 750	-.062						246	261	279	296	311	324	336*	347
2 000							247	262	280	297	312	325	337*	348
2 250								263	281	298	313	326	338*	349
2.500	+0							264	282	299	314	327	339*	350
2.750	-.094							265	283	300	315	328	340*	351
3 000								266	284	301	316	329	341*	352

* INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1976

(6) FOR CHANGE SEE SHEET 2.

REVIEWER NSA, IS, GL, EA, MU, EL, AV, MI
USER AT, OS, ME, PA, WY, YD, SM

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APPROVED 25 JUN 1953 REVISED (A) 5 APR 1962 (B) 18 JUN 1962 (C) 10 DEC 1964 (D) 18 AUG 1966 (E) 22 OCT 1970 (F) 2 FEB 76 (G) 28 JUL 1976

P.A. WC Other Cust AS 82	TITLE SCREW, MACHINE - 82° FLAT COUNTERSUNK HEAD, CROSS-RECESSED, CARBON STEEL, CADMIUM PLATED, UNF-2A	MILITARY STANDARD MS 35191
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES MS35189, MS35193, MS35195, MS35238, MS35240, MS35242, MS35244; and in part MS24583 (USAF), MS35358, MS35363, AN510	SHEET 1 OF 4

DD FORM SEP 67 672-1 (Coordinated)

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NOTES

1. MATERIAL Carbon steel in accordance with Fed. Std. No. 66 having minimum ultimate tensile strength of 60,000 PSI.
2. PROTECTIVE COATING. Cadmium plated in accordance with Type II, Class 3 of QQ-P-416 or MIL-C-81562.
3. THREADS The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H-28.
4. THREAD LENGTH: Screws 2.00 inches long and shorter shall have complete threads extending to within two threads of the head, or closer, if practicable Longer screws shall have a minimum thread length of 1.50 inches.
5. UNDERCUT HEADS. Screws above dotted lines shall have undercut heads
6. DIMENSIONS: All dimensions are in inches, unless otherwise specified.
7. RECESS: The recess shall be in accordance with MS9006.
8. PART NUMBER The MS part number consists of the MS number, plus the dash number. Example MS35191-209 plain screw or MS35191-209L self-locking screw.
9. Referenced documents shall be of the issue in effect on date of invitations for bids
10. For design feature purposes, this standard takes precedence over procurement documents referenced herein.
11. SELF-LOCKING ELEMENT:
    (a) The self-locking element shall be a patch type or longitudinal strip in accordance with MIL-F-18240.
    (G) (b) Maximum ring gage diameter that locking region of screw must pass thru freely or with finger pressure shall be the nominal size plus 0.010
    (c) For "X" and "Y" dimensions and design and usage limitations, see MS15981

REVIEWER NSA, IS, GL, EA, MU, EL, AV, MI
USER AT, OS, ME, PA, WY, YD, SH

APPROVED 25 JUN 1953 REVISED (A) 5 APR 1962 (B) 18 JUN 1962 (C) 10 DEC 1964 (D) 18 AUG 1966 (E) 22 OCT 1970 (F) 2 FEB 1976 (G) FOR CHANGE SEE SHEET 2.

TABLE II
INTERCHANGEABILITY

The screws covered by the following part numbers are cancelled after 5 April 1962. The cancelled screws cannot always replace the new screws and should be used until existing stocks are depleted. Use only the new screws for design and replacement. Replacement shall be in accordance with this table.

PART NUMBERS							
CANCELLED	NEW	CANCELLED	NEW	CANCELLED	NEW	CANCELLED	NEW
MS35189 MS35191 MS35193 MS35195	MS35191	MS35189 MS35191 MS35193 MS35195	MS35191	MS24583 MS35189 MS35191 MS35193 MS35195	MS35191	MS24583 MS35189 MS35191 MS35193 MS35195	MS35191
1	213	26	238	51	268	79	299
2	214	27	239	52	269	80	302
3	215	28	240	53	270	81	303
4	216	29	241	54	271	82	304
5	217	30	242	55	272	83	305
6	218	31	243	56	273	84	306
7	219	32	244	57	274	85	307
8	220	33	245	58	275	86	308
9	221	34	246	59	276	87	309
10	222	35	247	60	277	88	310
11	223	36	248	61	278	89	311
12	224	37	249	62	279	90	312
13	225	38	250	63	280	91	313
14	226	39	251	64	281	92	314
15	227	40	252	65	282	93	317
16	228	41	253	66	286	94	318
17	229	42	254	67	287	95	319
18	230	43	255	68	288	96	320
19	231	44	256	69	289	97	321
20	232	45	257	70	290	98	322
21	233	46	258	71	291	99	323
22	234	47	259	72	292	100	324
23	235	48	260	73	293	101	325
24	236	49	261	74	294	102	326
25	237	50	262	75	295	103	327
				76	296	104	328
				77	297	105	329
				78	298		

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TABLE II - CONT'D

PART NUMBERS

CANCELLED			NEW	CANCELLED			NEW	CANCELLED			NEW	CANC	NEW
MS35238 MS35240 MS35242 MS35244	AN510		MS35191	MS35238 MS35240 MS35242 MS35244	AN510		MS35191	MS35238 MS35240 MS35242 MS35244	AN510		MS35191	MS35238 MS35240 MS35242 MS35244	MS35191
1	0-2	0R2	201	34			234	68	10-4	10R4	268	102	302
2	0-3	0R3	202	35	6-4	6R4	235	69	10-5	10R5	269	103	303
3	0-4	0R4	203	36	6-5	6R5	236	70	10-6	10R6	270	104	304
4	0-5	0R5	204	37	6-6	6R6	237	71	10-7	10R7	271	105	305
5	0-6	0R6	205	38	6-7	6R7	238	72	10-8	10R8	272	106	306
	0-7	0R7	*	39	6-8	6R8	239		10-9	10R9	*	107	307
	0-8	0R8	*		6-9	6R9	*	73	10-10	10R10	273	108	308
6	1-2	1R2	206	40	6-10	6R10	240		10-11	10R11	*	109	309
7	1-3	1R3	207		6-11	6R11	*	74	10-12	10R12	274	110	310
8	1-4	1R4	208	41	6-12	6R12	241		10-13	10R13	*	111	311
9	1-5	1R5	209		6-13	6R13	*	75	10-14	10R14	275	112	312
10	1-6	1R6	210	42	6-14	6R14	242		10-15	10R15	*	113	313
11	1-7	1R7	211		6-15	6R15	*	76	10-16	10R16	276	114	314
12	1-8	1R8	212	43	6-16	6R16	243		10-18	10R18	*	115	315
13			213		6-18	6R18	*	77	10-20	10R20	277	116	316
14	2-3	2R3	214	44	6-20	6R20	244		10-22	10R22	*	117	317
15	2-4	2R4	215		6-22	6R22	*	78	10-24	10R24	278	118	318
16	2-5	2R5	216	45	6-24	6R24	245		10-26	10R26	*	119	319
17	2-6	2R6	217		6-26	6R26	*	79	10-28	10R28	279	120	320
18	2-7	2R7	218	46	6-28	6R28	246		10-30	10R30	*	121	321
19	2-8	2R8	219		6-30	6R30	*	80	10-32	10R32	280	122	322
	2-9	2R9	*	47	6-32	6R32	247		10-34	10R34	*	123	323
20	2-10	2R10	220		6-34	6R34	*	81	10-36	10R36	281	124	324
	2-11	2R11	*		6-36	6R36	*		10-38	10R38	*	125	325
21	2-12	2R12	221		6-38	6R38	*	82	10-40	10R40	282	126	326
	3-3	3R3	*		6-40	6R40	*		10-42	10R42	*	127	327
	THRU	THRU	*	48			248	83	10-44	10R44	283	128	328
	3-12	3R12	*	49			249		10-46	10R46	*	129	329
22			222	50	8-4	8R4	250	84	10-48	10R48	284	130	330
23	4-3	4R3	223	51	8-5	8R5	251	85			285	131	331
24	4-4	4R4	224	52	8-6	8R6	252	86			286	132	332
25	4-5	4R5	225	53	8-7	8R7	253	87	416-6	416R6	287	133	333
26	4-6	4R6	226	54	8-8	8R8	254	88	416-7	416R7	288	134	334
27	4-7	4R7	227		8-9	8R9	*	89	416-8	416R8	289	135	335
28	4-8	4R8	228	55	8-10	8R10	255		416-9	416R9	*	136	336
	4-9	4R9	*		8-11	8R11	*	90	416-10	416R10	290	137	337
29	4-10	4R10	229	56	8-12	8R12	256		416-11	416R11	*	138	338
	4-11	4R11	*		8-13	8R13	*	91	416-12	416R12	291	139	339
30	4-12	4R12	230	57	8-14	8R14	257		416-13	416R13	*	140	340
	4-13	4R13	*		8-15	8R15	*	92	416-14	416R14	292	141	341
31	4-14	4R14	231	58	8-16	8R16	258		416-15	416R15	*	142	342
	4-15	4R15	*		8-18	8R18	*	93	416-16	416R16	293	143	343
32	4-16	4R16	232	59	8-20	8R20	259		416-18	416R18	*	144	344
	4-18	4R18	*		8-22	8R22	*	94	416-20	416R20	294	145	345
	4-20	4R20	*	60	8-24	8R24	260		416-22	416R22	*	146	346
	4-22	4R22	*		8-26	8R26	*	95	416-24	416R24	295	147	347
	4-24	4R24	*	61	8-28	8R28	261		416-26	416R26	*	148	348
	4-26	4R26	*		8-30	8R30	*	96	416-28	416R28	296	149	349
	4-28	4R28	*	62	8-32	8R32	262		416-30	416R30	*	150	350
	5-4	5R4	*		8-34	8R34	*	97	416-32	416R32	297	151	351
	THRU	THRU	*	63	8-36	8R36	263		416-34	416R34	*	152	352
	5-16	5R16	*		8-38	8R38	*	98	416-36	416R36	298	153	*
	5-18	5R18	*	64	8-40	8R40	264		416-38	416R38	*	THRU	*
	5-20	5R20	*		8-42	8R42	*	99	416-40	416R40	299	181	*
	5-22	5R22	*	65	8-44	8R44	265		416-42	416R42	*		
	5-24	5R24	*		8-46	8R46	*	100	416-44	416R44	300		
	5-26	5R26	*	66	8-48	8R48	266		416-46	416R46	*		
	5-28	5R28	*	67			267	101	416-48	416R48	301		
33			233										

* No Replacement

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P.A.	WC
Other Cust	AS 82
PROCUREMENT SPECIFICATION	FF-S-92

TITLE: SCREW, MACHINE - 82° FLAT COUNTERSUNK HEAD, CROSS-RECESSED, CARBON STEEL, CADMIUM PLATED, UNF-2A

SUPERSEDES MS35189, MS35193, MS35195, MS35238, MS35240, MS35242, MS35244; and in part MS24583 (USAF), MS35358, MS35363, AN510

MILITARY STANDARD

MS 35191

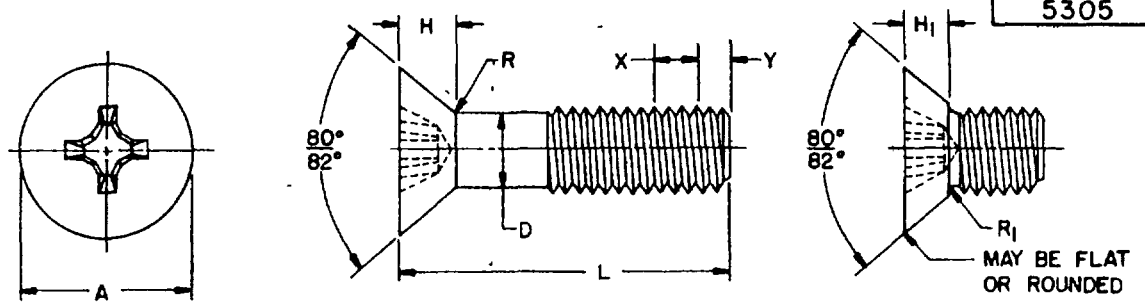
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FULL HEAD — UNDERCUT HEAD

TABLE III
METRIC TRANSLATION OF TABLE I

	NOMINAL SIZE	MM	1.524	1.854	2.184	2 845	3 505	4 166	4.826	6 350	7.938	9.525	11.112	12.700
	PITCH		.32	.35	.40	.53	64	71	79	.91	1.06	1.06	1 27	1.27
D	BODY DIA	MAX	1.52	1.85	2 18	2 85	3.51	4 17	4.83	6.35	7.94	9.53	11.11	12 70
		MIN	1.26	1.56	1.86	2.42	3.01	3.62	4.21	5.65	7.13	8.71	10.15	11.73
A	HEAD DIA	MAX SHARP	3.02	3.71	4.37	5.72	7.09	8.43	9.78	12 88	16.13	19.36	20.63	22.23
		MIN SHARP	2.67	3.30	3.96	5.26	6.53	7.82	9.12	12.12	15.24	18.34	19 48	21.11
		ABS MIN	2.52	3.12	3.73	4.92	6.20	7 42	8.64	11.48	14.43	17.40	18.36	19.69
H	HEAD HEIGHT	REF	.89	1.09	1.30	1.70	2.11	2.54	2.95	3 89	4.85	5.84	5 66	5 66
H_1	HEAD HEIGHT UNDERCUT	MAX	.64	.79	.91	1.19	1.50	1 78	2.06	2.72	3.40	4.09	3.96	3.96
		MIN	.46	.58	.71	.97	1.22	1.47	1.73	2.34	2.95	3.56	3 38	3.30
R	RADIUS	MAX	.61	.74	.86	1.14	1.40	1.68	1.93	2 54	3.18	3.81	4.45	5 08
R_1	RADIUS-UNDERCUT	MAX	.23	.28	.33	.43	.53	.64	.74	.97	1.19	1.42	1.68	1.91
	TENSILE STRENGTH KN - MIN		.48	.73	1.05	1.76	2.69	3 92	5.34	9.70	15.5	23.4	31.7	42.7
L	LENGTH	TOL	DASH NO.	DASH NO.	DASH NO.	DASH NO	DASH NO.	DASH NO	DASH NO.	DASH NO.	DASH NO	DASH NO.	DASH NO.	DASH NO.
	3.18		201	206*	213	222	233	248						
	4.78		202	207*	214	223	234	249	267					
	6.35		203	208*	215	224	235	250	268	285				
	7.93		204	209*	216	225	236	251	269	286				
	9.53	+0	205	210*	217	226	237	252	270	287	302			
	11.13	-.79		211*	218	227	238	253	271	288	303			
	12.70			212*	219	228	239	254	272	289	304	317		
	15.88				220	229	240	255	273	290	305	318	330*	
	19.05				221	230	241	256	274	291	306	319	331*	342
	22 23					231	242	257	275	292	307	320	332*	343
	25.40					232	243	258	276	293	308	321	333*	344
	31.75						244	259	277	294	309	322	334*	345
	38.10	+0					245	260	278	295	310	323	335*	346
	44.45	-1.58					246	261	279	296	311	324	336*	347
	50.80						247	262	280	297	312	325	337*	348
	57.15							263	281	298	313	326	338*	349
	63.50	+0						264	282	299	314	327	339*	350
	69.85	-2.39						265	283	300	315	328	340*	351
	76.20							266	284	301	316	329	341*	352

NOTE: All dimensions are in millimeters (MM), unless otherwise specified.

* INACTIVE FOR NEW DESIGN AFTER 2 FEBRUARY 1976

REVIEWER: EL, NSA, IS, GL, EA, MU, AV, MI
USER: AT, OS, ME, PA, WV, YD, SN

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